Generalizing the Phase-Diversity Algorithm to Nonlinear Excitation

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1 Derivation of the Gradient

The goal will be to understand the effects of nonlinear excitation:

$$s[\phi] = |\mathcal{F}(pe^{i\phi})|^{2n}$$

where $n \in \mathbb{N}$ is 1 for one-photon excitation, 2, for two-photon excitation, etc. We begin by writing the cost functional, with the minimzer $\hat{F}(\mathbf{u})$ already plugged in:

$$J[\phi] = -\frac{1}{2}\iint_{\mathbb{R}^2}\left[\frac{|\sum_{k=1}^{K} D_k^* S_k[\phi]|^2}{\gamma + \sum_{k=1}^{K}|S_k[\phi]|^2}\right] d\mathbf{u} + \text{Constant} + \text{Regularization} \tag{1}$$

We define, for convenience:

$$P[\phi] = \sum_{k=1}^{K} D_k^* S_k[\phi] \tag{2}$$

$$Q[\phi] = \gamma + \sum_{k=1}^{K} S_k^*[\phi] S_k[\phi] \tag{3}$$

Which allows us to rewrite the cost functional in Equation 1 as follows:

$$J[\phi] = -\frac{1}{2}\iint_{\mathbb{R}^2}\frac{P[\phi]P^*[\phi]}{Q[\phi]} \tag{4}$$

We will attempt to apply the Gateaux derivative in the direction ψ:

$$\delta_\psi J[\phi] = \lim_{\tau\to 0}\frac{J[\phi+\tau\psi] - J[\phi]}{\tau} = \frac{d}{d\tau}J[\phi+\tau\psi]|_{\tau=0} \tag{5}$$

We start with:

$$\delta_\psi J[\phi] = \delta_\psi\left(-\frac{1}{2}\iint_{\mathbb{R}^2}\frac{P[\phi]P^*[\phi]}{Q[\phi]}\right) \tag{6}$$

Using the quotient rule, we get:

$$\begin{aligned}
d_\psi \iint \frac{P[\phi]P^*[\phi]}{Q[\phi]} &= \iint \frac{Q[\phi]\delta_\psi(P[\phi]P^*[\phi]) - P[\phi]P^*[\phi]\delta_\psi(Q[\phi])}{Q[\phi]^2} \\
&= \iint \frac{Q\left[P^*\sum_{k=1}^{K} D_k^*\delta_\psi S_k + P\sum_{k=1}^{K} D_k\delta_\psi S_k^*\right] - PP^*\sum_{k=1}^{K}(S_k^*\delta_\psi S_k + S_k\delta_\psi S_k^*)}{Q^2} \\
&= \sum_{k=1}^{K}\iint \frac{P^*QD_k^* - PP^*S_k^*}{Q^2}\delta_\psi S_k + \text{c.c} \\
&= \sum_{k=1}^{K}\langle \delta_\psi S_k[\phi], V_k\rangle + \text{c.c}
\end{aligned} \tag{7}$$

Where c.c denotes the complex conjugate of the preceeding term, and V_k is define as:

$$V_k = \frac{PQ^* D_k - |P|^2 S_k}{Q^2} = F^* D_k - |F|^2 S_k \tag{8}$$

Now we expand each of these terms, beginning with Parseval's theorem:

$$\begin{aligned} \langle \delta_\psi S_k[\phi], V_k \rangle &= \langle \delta_\psi s_k, v_k \rangle \\ &= \langle \delta_\psi |h_k|^{2n}, v_k \rangle \\ &= \langle n|h_k|^{2n-2}(h_k^* \delta_\psi h_k + h_k \delta_\psi h_k^*), v_k \rangle \end{aligned} \tag{9}$$

Adding the complex conjugate, we get that each term of the summation in Equation 7 is:

$$\begin{aligned} \langle \delta_\psi S_k, V_k \rangle + \langle V_k, \delta_\psi S_k \rangle &= \langle n|h_k|^{2n-2}(h_k^* \delta_\psi h_k + h_k \delta_\psi h_k^*), v_k \rangle + \langle v_k, n|h_k|^{2n-2}(h_k^* \delta_\psi h_k + h_k \delta_\psi h_k^*) \rangle \\ &= n\langle |h_k|^{2n-2}(h_k^* \delta_\psi h_k + h_k \delta_\psi h_k^*), v_k \rangle + n\langle |h_k|^{2n-2}(h_k \delta_\psi h_k^* + h_k^* \delta_\psi h_k), v_k^* \rangle \\ &= n\langle |h_k|^{2n-2}(h_k^* \delta_\psi h_k + h_k \delta_\psi h_k^*), 2\Re(v_k) \rangle \\ &= 2n\langle 2\Re(h_k^* \delta_\psi h_k), |h_k|^{2n-2}\Re(v_k) \rangle \\ &= 4n\Re\left(\langle h_k^* \delta_\psi h_k, |h_k|^{2n-2}\Re(v_k) \rangle\right) \\ &= 4n\Re\left(\langle \delta_\psi h_k, |h_k|^{2n-2} h_k \Re(v_k) \rangle\right) \\ &= 4n\Re\left(\langle \delta_\psi h_k, r_k \rangle\right) \end{aligned} \tag{10}$$

Where r_k is defined as

$$r_k = |h_k|^{2n-2} h_k \Re(v_k) \tag{11}$$

Now, we note that $h_k = \mathcal{F}^{-1}(H_k)$, where $\delta_\psi H_k = iH_k\psi$. This gives:

$$\langle \delta_\psi S_k, V_k \rangle + \langle V_k, \delta_\psi S_k \rangle = 4n\Re\left(\langle \mathcal{F}^{-1}(iH_k\psi), r_k \rangle\right) \tag{12}$$

Applying Parseval's again, we get:

$$\begin{aligned} \langle \delta_\psi S_k, V_k \rangle + \langle V_k, \delta_\psi S_k \rangle &= 4n\Re\left(\langle iH_k\psi, \mathcal{F}(r_k) \rangle\right) \\ &= 4n\Re(i\langle H_k\psi, \mathcal{F}(r_k) \rangle) \\ &= -4n\Im(\langle H_k\psi, \mathcal{F}(r_k) \rangle) \\ &= -4n\Im(\langle \psi, H_k^* \mathcal{F}(r_k) \rangle) \\ &= 4n\langle \psi, \Im\left(H_k^* \mathcal{F}(r_k)\right) \rangle \end{aligned} \tag{13}$$

$$\tag{14}$$

where the last move is allowed since ψ is completely real. Plugging back in to Equation 6, we get:

$$\begin{aligned} \delta_\psi J[\phi] &= -\frac{1}{2} \sum_{k=1}^{K} 4n\langle \psi, \Im\left(H_k^* \mathcal{F}(r_k)\right) \rangle \\ &= \langle \psi, -2n \sum_{k=1}^{K} \Im(H_k^* \mathcal{F}(r_k)) \rangle \end{aligned} \tag{15}$$

Now, we note the definition of the gradient:

$$\delta_\psi J[\phi] = \langle \psi, \nabla J[\phi] \rangle \tag{16}$$

Inspecting Equation 15, we conclude:

$$\nabla J[\phi] = -2n \sum_{k=1}^{K} \Im(H_k^* \mathcal{F}(r_k)) \tag{17}$$

$$= -2n \sum_{k=1}^{K} \Im(H_k^* \mathcal{F}(|h_k|^{2n-2} h_k \Re(\mathcal{F}^{-1}(V_k)))) \tag{18}$$

From here on, we will refer to this gradient as $g[\phi]$.

2 Derivation of the Hessian

We will begin with the assumption that the phase ϕ can be expanded over a basis $\{ Z_i \}$:

$$\phi(\mathbf{x}) = \sum_i \langle \phi, Z_i \rangle Z_i(x) \tag{19}$$

The goal now is to find the discrete Hessian. If there are m basis functions in our expansion, we write its i, jth entry as:

$$\mathbf{H}_{ij} = \langle H[\phi] Z_j, Z_i \rangle, \quad 1 \leq i, j \leq m \tag{20}$$

Where the Hessian entries can be approximated by the finite differences of the gradient:

$$H[\phi] Z_j \approx \frac{g[\phi + \tau Z_j] - g[\phi]}{\tau}, \qquad \tau \text{ small} \tag{21}$$

Now, we will attempt to find the Gauss-Newton approximation of $H[\phi]Z_j$ (the action of the Hessian on the jth basis function). Begin by rewriting the cost functional in Equation 1 as follows:

$$J[\phi] = \frac{1}{2} \sum_{k=1}^{K} \sum_{j=1}^{k-1} ||R_{jk}[\phi]||^2 + \frac{\gamma}{2} ||D^{1/2}[\phi]||^2 \tag{22}$$

Where we define:

$$R_{jk}[\phi] = \frac{D_k S_j[\phi] - D_j S_k[\phi]}{Q^{1/2}[\phi]} \tag{23}$$

$$D[\phi] = \frac{\sum_{k=1}^{K} |D_k|^2}{Q[\phi]} \tag{24}$$

$$Q[\phi] = \gamma + \sum_{k=1}^{K} S_k^*[\phi] S_k[\phi] \tag{25}$$

We begin by dropping the term on the RHS of Equation 22. This gives us a simple least-squares cost function, which allows us to write the Gauss-Newton approximation as follows:

$$\langle H_{GN}[\phi] Z, \psi \rangle \approx \sum_{k=1}^{K} \sum_{j=1}^{k-1} \langle R'_{jk}[\phi] Z, R'_{jk}[\phi] \psi \rangle \tag{26}$$

It is helpful now to redefine some variables:

$$\tilde{D}_k = \frac{D_k}{Q^{1/2}} \tag{27}$$

$$\tilde{R}_{jk}[\phi] = \tilde{D}_k S_j[\phi] - \tilde{D}_j S_k[\phi] \tag{28}$$

$$U_{jk} = \tilde{R}'_{jk}[\phi] = \tilde{D}_k S'_j[\phi] Z - \tilde{D}_j S'_k[\phi] Z \tag{29}$$

Which allows us to write:

$$\langle H_{GN}[\phi] Z, \psi \rangle = \sum_{k=1}^{K} \sum_{j=1}^{k-1} \langle U_{jk}, \tilde{R}'_{jk}[\phi] \psi \rangle \tag{30}$$

Now we approximate $S'_j[\phi]Z$ for fixed $Z \in L^2(\mathbb{R}^2)$. We drop the subscript j and let W be arbitrary. Them, we get:

$$\begin{aligned} \langle S'[\phi] Z, W \rangle &= \langle \delta_Z S[\phi], W \rangle \\ &= \langle \delta_Z s[\phi], w \rangle \\ &= \langle \delta_Z |h|^{2n}, w \rangle \\ &= \langle n|h|^{2n-2} (h^* \delta_Z h + h \delta_Z h^*), w \rangle \end{aligned} \tag{31}$$

Now, we again note that $h = \mathcal{F}^{-1}(H)$ where $\delta_Z H = iHZ$, which gives $\delta_Z h = \mathcal{F}^{-1}(iHZ)$ This gives:

$$\begin{aligned}\langle S'[\phi]Z, W\rangle &= n\langle i|h|^{2n-2}h^*\mathcal{F}^{-1}(HZ), w\rangle + n\langle -i|h|^{2n-2}h\left(\mathcal{F}^{-1}(HZ)\right)^*, w\rangle \\ &= \langle -2n\Im(|h|^{2n-2}h^*\mathcal{F}^{-1}(HZ)), w\rangle\end{aligned} \tag{32}$$

Which allows us to identify:

$$S'[\phi]Z = -2n\mathcal{F}(\Im(|h|^{2n-2}h^*\mathcal{F}^{-1}(HZ))) \tag{33}$$

Giving:

$$U_{jk} = -2n\tilde{D}_k\mathcal{F}(\Im(|h_j|^{2n-2}h_j^*\mathcal{F}^{-1}(H_jZ))) + 2n\tilde{D}_j\mathcal{F}(\Im(|h_k|^{2n-2}h_k^*\mathcal{F}^{-1}(H_kZ))) \tag{34}$$

Now we define:

$$W_{jk,i} = \tilde{D}_i^* U_{jk} \tag{35}$$

$$w_{jk,i} = \mathcal{F}^{-1}(\tilde{D}_i^* U_{jk}) \tag{36}$$

$$\tag{37}$$

Now, examining Equation 30, we unpack each of the terms.

$$\langle U_{jk}, \tilde{D}_k S_j'[\phi]\psi\rangle = \langle S_j'[\phi]^* W_{jk,k}, \psi\rangle \tag{38}$$

$$= \langle 2n\Im\left(H_j^*\mathcal{F}(|h_j|^{2n-2}h_j\mathcal{F}^{-1}(W_{jk,k}))\right), \psi\rangle \tag{39}$$

Looking back at Equation 30, we can identify:

$$H_{GN}[\phi]Z = 2n\sum_{k=1}^{K}\sum_{j=1}^{k-1}\Im\left[H_j^*\mathcal{F}\left(|h_j|^{2n-2}h_j\mathcal{F}^{-1}(\tilde{D}_k^*U_{jk})\right)\right] - \Im\left[H_k^*\mathcal{F}\left(|h_k|^{2n-2}h_k\mathcal{F}^{-1}(\tilde{D}_j^*U_{jk})\right)\right] \tag{40}$$

To clean this up and match the convention from the original paper, we redefine:

$$\tilde{U}_{jk} = \tilde{D}_j\mathcal{F}(\Im(|h_k|^{2n-2}h_k^*\mathcal{F}^{-1}(H_kZ))) - \tilde{D}_k\mathcal{F}(\Im(|h_j|^{2n-2}h_j^*\mathcal{F}^{-1}(H_jZ))) \tag{41}$$

Which gives:

$$H_{GN}[\phi]Z = 4n^2\sum_{k=1}^{K}\sum_{j=1}^{k-1}\Im\left[H_j^*\mathcal{F}\left(|h_j|^{2n-2}h_j\mathcal{F}^{-1}(\tilde{D}_k^*\tilde{U}_{jk})\right)\right] - \Im\left[H_k^*\mathcal{F}\left(|h_k|^{2n-2}h_k\mathcal{F}^{-1}(\tilde{D}_j^*\tilde{U}_{jk})\right)\right] \tag{42}$$